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Other Ways

Alternatives to Many-Worlds

David Albert, now a philosophy professor at Columbia and one of the world's leading researchers in the foundations of quantum mechanics, had a very typical experience as a graduate student who became interested in quantum foundations. He was in the PhD program in the physics department at Rockefeller University when, after reading a book by eighteenth-century philosopher David Hume on the relationship of knowledge and experience, he came to believe that what physics lacked was a good understanding of the quantum measurement problem. (Hume didn't know about the measurement problem, but Albert connected dots in his head.) Nobody at Rockefeller in the late 1970s was interested in thinking along those lines, so Albert struck up a long-distance collaboration with the famous Israeli physicist Yakir Aharonov, resulting in several influential papers. But when he suggested submitting that work for his PhD thesis, the powers that be at Rockefeller were aghast. Under penalty of being kicked out of the program entirely, Albert was forced to write a separate thesis in mathematical physics. It was, as he recalled, "clearly being assigned because it was thought it would be good for my character. There was an explicitly punitive element there."

Physicists have been very bad at coming to consensus about what the foundations of quantum mechanics actually are. But in the second half of the twentieth century, they did come to a remarkable degree of consensus on a related issue: whatever the foundations of quantum mechanics are, we certainly shouldn't *talk* about them. Not while there was real work to be